

L25 — Infix to Postfix Conversion

Unit: 2 | Chapter: Stacks & Recursion | BT Level: BT5 | CO: CO3

Learning Objectives

- Define operator precedence and associativity rules
 - Implement the Shunting-Yard algorithm for infix to postfix conversion
 - Trace the conversion for complex expressions with parentheses
 - Handle all standard operators including exponentiation
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1. Why Convert Infix to Postfix?

Infix: $3 + 4 * 2 \rightarrow$ requires knowing precedence and parentheses

Postfix: $3\ 4\ 2\ *\ + \rightarrow$ machine-evaluable left-to-right with a simple stack

Compilers convert infix arithmetic expressions to postfix (or prefix) internally before generating code. This is a core step in **expression parsing**.

2. Operator Precedence and Associativity

Operator	Precedence	Associativity
$^$ (exponent)	4 (highest)	Right-to-Left
$*$, $/$, $\%$	3	Left-to-Right
$+$, $-$	2	Left-to-Right
$($	1 (lowest when in stack) —	

Rules:

- Higher precedence operators are applied first
 - Left-to-right: $a - b - c = (a - b) - c$
 - Right-to-left: $a ^ b ^ c = a ^ (b ^ c)$
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3. Shunting-Yard Algorithm (Dijkstra, 1961)

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Algorithm infix_to_postfix(expression):  
    output = []           # Output queue (postfix result)  
    operator_stack = []  
  
    for each token in expression:  
        if token is a number/variable:  
            append to output  
  
            elif token is '(':
```

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        push '(' onto operator_stack

    elif token is ')':
        while top of stack != '(':
            pop from stack and append to output
        pop '(' from stack (discard it)

    elif token is an operator:
        while (operator_stack not empty AND
              top != '(' AND
              (top has higher precedence than token OR
               (same precedence AND token is left-associative))):
            pop from stack and append to output
        push token onto operator_stack

while operator_stack not empty:
    pop from stack and append to output

return output

```

4. Implementation

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def infix_to_postfix(expression):
    precedence = {'+': 2, '-': 2, '*': 3, '/': 3, '%': 3, '^': 4}
    right_assoc = {'^'}    # Right-associative operators

    output = []
    op_stack = []
    tokens = expression.split()

    for token in tokens:
        if token not in precedence and token not in '()':
            # Operand (number or variable)
            output.append(token)

        elif token == '(':
            op_stack.append(token)

        elif token == ')':
            while op_stack and op_stack[-1] != '(':
                output.append(op_stack.pop())
            if not op_stack:
                raise ValueError("Mismatched parentheses")
            op_stack.pop()    # Remove '('

        else:
            # Operator
            while (op_stack and
                  op_stack[-1] != '(' and
                  op_stack[-1] in precedence and
                  (precedence[op_stack[-1]] > precedence[token] or

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        (precedence[op_stack[-1]] == precedence[token] and
         token not in right_assoc)):
            output.append(op_stack.pop())
            op_stack.append(token)

while op_stack:
    if op_stack[-1] == '(':
        raise ValueError("Mismatched parentheses")
    output.append(op_stack.pop())

return ' '.join(output)

```

5. Detailed Traces

Example 1: $A + B * C$

Token	Action	Stack	Output
A	Operand	[]	[A]
+	Push (empty stack)	[+]	[A]
B	Operand	[+]	[A, B]
*	$\text{prec}(*) > \text{prec}(+)$, push	[+, *]	[A, B]
C	Operand	[+, *]	[A, B, C]
END	Pop all: *, then +	[]	[A, B, C, *, +]

Result: $A \ B \ C \ * \ +$

Verify: evaluate $A \ B \ C \ * \ + \rightarrow A + (B * C) = A + B * C \checkmark$

Example 2: $(A + B) * C$

Token	Action	Stack	Output
(	Push	[(]	[]
A	Operand	[(]	[A]
+	Push (top is '(')	[(, +]	[A]
B	Operand	[(, +]	[A, B]
)	Pop until '(': pop +	[]	[A, B, +]
*	Push	[*]	[A, B, +]
C	Operand	[*]	[A, B, +, C]
END	Pop *:	[]	[A, B, +, C, *]

Result: $A \ B \ + \ C \ *$

Verify: $(A + B) * C \checkmark$

Example 3: $A \wedge B \wedge C$ (right-associative)

Token	Action	Stack	Output
A	Operand	[]	[A]
^	Push	[^]	[A]
B	Operand	[^]	[A, B]
^	Right-assoc: same prec, don't pop → push	[^, ^]	[A, B]
C	Operand	[^, ^]	[A, B, C]
END	Pop ^, ^:	[]	[A, B, C, ^, ^]

Result: A B C ^ ^

Verify: A ^ (B ^ C) (right-to-left associativity) ✓

Example 4: 3 + 4 * 2 / (1 - 5) ^ 2 ^ 3

Token	Stack	Output
3	[]	[3]
+	[+]	[3]
4	[+]	[3,4]
*	[+,*]	[3,4]
2	[+,*]	[3,4,2]
/	pop * → [+], push / → [+,/]	[3,4,2,*]
(	[+/, (]	[3,4,2,*]
1	[+/, (]	[3,4,2,*,1]
-	[+/, (-]	[3,4,2,*,1]
5	[+/, (-]	[3,4,2,*,1,5]
)	pop -: [+,/]	[3,4,2,*,1,5,-]
^	[+/, ^]	[3,4,2,*,1,5,-]
2	[+/, ^]	[3,4,2,*,1,5,-,2]
^	right-assoc: [+,/, ^, ^]	[3,4,2,*,1,5,-,2]
3	[+/, ^, ^]	[3,4,2,*,1,5,-,2,3]
END	pop ^, ^, /, +	[3,4,2,*,1,5,-,2,3,^,^,/,+]

Result: 3 4 2 * 1 5 - 2 3 ^ ^ / +

6. Complexity

Metric	Value
Time	O(n) — each token processed at most twice (push + pop)
Space	O(n) — stack holds at most n/2 operators

Summary

- Infix $\rightarrow$ Postfix conversion uses a stack to handle precedence and parentheses.
 - **Shunting-Yard algorithm:** push operands directly, pop operators with higher/equal precedence before pushing current.
 - Parentheses (and) control grouping — they never appear in postfix output.
 - Right-associative operators (like $\wedge$) are not popped when equal precedence matches.
 - Time and space: $O(n)$.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 10.1 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 3.4 (Springer, 2020)
- Dijkstra, "Algol 60 Translation" (1961) — original Shunting-Yard paper